strategy includes product and technical testing of components, subassemblies and the finished product. It correlates defects to optimise overall performance of the device and desired output for each stage. The system is divided into small blocks to test specified requirements for each block in the system. A risk-based approach validates hardware, software and peripheral devices' testing associated with the intended use of the system.

Cigniti Technologies, a software testing company, follows a five-step testing approach (Fig. 1), which is aligned with the development of product lifecycle, enabling a better understanding of test requirements.

Software validation and verification

End-to-end testing, certification and evaluation help ensure the conformity of the electrical or electronic equipment's safety and other essential performance requirements for global product approvals. Lynne A. Dunbrack, research vice president, IDC Health Insights, says, "If not properly secured, medical devices, such as bedside telemetry or implantable devices like pacemakers and insulin pumps, can introduce malware to the network." To mitigate these problems, software validation testing is required, which includes the following:

Interoperability of devices. There is a need for rigorous testing of devices and applications that interoperate and connect to deliver the required experience to the user. This ensures data security and privacy through a quality assurance plan. Tests are recorded and repeated for audit and compliance purposes. For example, if a wearable and its applications do not work in sync, it might not show accurate results. With innovations around Internet of Things (IoT) and smart devices, a need arises for devices, software control and application to integrate with the core software.

Functional validation of software. An application that helps patients or doctors to generate or access reports with a secure log-in must be functional all the time to deliver the required service accurately.

Security of applications. Security testing helps make applications hack-proof and sustainable in the challenging digital scenario. Validation and authentication of user log-ins, testing against firewalls and encrypting user data are some key measures. Healthcare applications or products need to adhere to the stringent Health Insurance Portability and Accountability Act requirements that ensure protection of patient health information.

Regulatory requirements for medical equipment

IEC 60601-xx standard family is the base for the approval procedure of medical electrical equipment in most regulatory frameworks worldwide. Product safety requirements for electrical active medical devices are documented and internationally harmonised under this standard. It is a series of technical standards that ensure the safety of medical electrical equipment.

IEC 60601-1 edition 3 is the internationally-harmonised safety standard used for all electronic medical devices' safety evaluation. According to this, medical devices are evaluated against electrical shocks, mechanical hazards, unwanted and excessive radiation, ignition hazards, abnormal operation, and faults and constructional defects.

IEC 60601-1 edition 3.1 ensures that electrical, mechanical or functional failures shall not pose any risk to patients and operators. It is recognised as a prerequisite for the commercialisation of electrical medical equipment in many countries. It is the newest published general standard with around 1500 single specific requirements.

IEC 60601-1 edition 2.0 includes requirements for the construction of transformers. Lithium batteries must comply with IEC 60086-4 for primary cells and IEC 62133 for secondary cells. IEC 60601-1 edition 2.0 also includes humidity testing requirements.

IEC 60601-1-2 defines essential performance parameters for medical

Fig. 2: UniPulse 400 (Credit: www.rigelmedical.com)

equipment with regards to emissions and immunity to electromagnetic interference (EMI) emitted by medical equipment and systems. IEC 60601-1-2 edition 4 introduced significant technical revisions such as risk analysis and management. Now, it includes specifications for immunity test levels according to the environments of intended use, that too dependent on locations that are harmonised with IEC 60601-1-11: hospital and home healthcare facility environment, and special environments.

IEC 17025 includes environmental testing for different climatic conditions and mechanical stress that products are exposed to during their lifetime. This exposes weaknesses in product design or performance that could occur during operation, particularly at extreme levels. It also helps with compliance of products with international regulations, for easier to access global markets.

ISO 14708-3 verifies that electrical and environmental impacts do not affect implantable devices. It includes electrical safety and electromagnetic compatibility (EMC) testing for active implantable medical devices, implants for surgery and implantable neurostimulators.

ISO 14971 checks the compliance of risk management of medical devices. It is meant to identify, quantify and mitigate risks that may be present in all settings of intended use. It foresees misuse and conditions of products or systems. Manufactur-